



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA0509

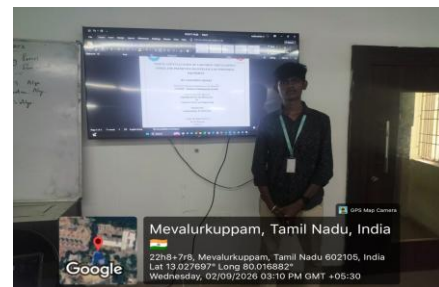
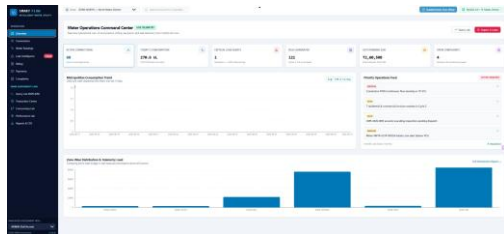
Slot: A

Course Name: Database Management Systems

Course Faculty: Dr. R. Kesavan

Project Title: SMARTFLOW: Intelligent Water Utility Management System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: SMARTFLOW – Intelligent Water Utility Management System is a database-driven application developed using **MySQL 8.0** to manage water consumption, billing, payments, leak detection, and customer complaints. The system stores large-scale **hourly meter readings** and uses **B+Tree indexing and a custom hashing technique** for efficient data retrieval. The system implements **query optimization using MySQL EXPLAIN**, along with **ACID transactions, SAVEPOINT, row-level locking, and Optimistic Concurrency Control (OCC)** to maintain data consistency and prevent duplicate billing and lost updates. A **Leak Intelligence module** identifies abnormal consumption patterns and classifies potential leakage risks. SMARTFLOW provides a **React-based web interface** with modules for dashboard monitoring, meter readings, leak detection, billing, payments, complaints, query analysis, transactions, concurrency testing, and performance benchmarking. The project demonstrates the practical application of core **Database Management System concepts** in a scalable water utility environment.

Student Signature

Guide Signature